

EDUCATION

- **University of California, Davis, USA** GPA: 4.0/4.0
• *Doctor of Philosophy (Ph.D.) in Computer Science* September 2023 – Present
 - **Advisor:** Dr. Aditya V. Thakur
- **Indian Institute of Science (IISc), Bengaluru, India** GPA: 8.0/10
• *Master of Technology (M.Tech.) in Computer Science and Engineering* August 2021 – July 2023
 - **Advisor:** Dr. Deepak D'Souza
- **Jadavpur University, Kolkata, India** GPA: 9.74/10
• *Bachelor of Engineering (B.E.) in Information Technology* July 2017 – June 2021

RESEARCH INTERESTS

Formal Methods; Machine Learning (Generative AI) for Systems and Software Engineering; High-Performance Computing and Compiler Optimizations.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Fortran, Java, R, MATLAB
- **Database:** MySQL, Oracle 10g, HDF5
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, NumPy, OpenCV, ONNX, TensorRT
- **HPC & Parallelism:** MPI, OpenMP, CUDA, cuDNN, cuBLAS
- **Formal Methods:** LLVM/Clang, ROSE, LFortran, Soot, Angr, Alloy, Spin, Rodin, VCC, JPF, AFL, KLEE, Z3, Frama-C, TVM, Gurobi, Infer, dReal, Daikon, DIG, Wolfram
- **Software & Dev:** HTML, Javascript, CSS, AJAX, XML, LaTeX, Moea Framework, Android Studio

PUBLICATIONS

- **A. Das, M. Wasif Ansari and R. Basak**, “Covid-19 Face Mask Detection Using TensorFlow, Keras and OpenCV”, 2020 IEEE 17th India Council International Conference (INDICON), New Delhi, India, 2020, pp. 1–5, doi: [10.1109/INDICON49873.2020.9342585](https://doi.org/10.1109/INDICON49873.2020.9342585). [Citation Count: 224 (as of August 2026)]

PROFESSIONAL SERVICE

- Artifact Evaluation Committee (AEC) Member — **ISSTA 2026** and **ASE 2026**.

EXPERIENCE

- **Lawrence Livermore National Laboratory (LLNL)** Livermore, CA
• *Graduate Computing Intern* June 2026 – September 2026
 - **Contribution:** Implemented a new branch-flip detection feature in the tool [FPChecker](#) by extending its LLVM instrumentation and runtime to identify floating-point branch conditions that may become unstable due to accumulated numerical error.
- **University of California, Davis** Davis, CA
• *Teaching Assistant – Programming Languages (ECS 140A, Undergraduate Level)* Winter Quarters 2024 & 2025
- **Indian Institute of Science (IISc), Bengaluru** Bengaluru, India
• *Teaching Assistant – Program Analysis and Verification (E0 227, Graduate Level)* Fall Semester 2022
- **Indian Institute of Technology (IIT) Guwahati** Guwahati, India
• *Summer Intern (Data Science) – Joint Work with Duke-NUS Medical School, Singapore* May 2020 – July 2020
 - **Developed:** an R-based [interactive web app](#) for state-wise COVID-19 prediction and interactive data visualization.
 - **Contribution:** Designed statistical pipelines, dashboards, and collaborated on model validation with researchers.
 - **Impact:** Improved access to public health data during the pandemic, with the app receiving public recognition from India's Ministry of Education.

PROJECTS

- **Code-aware Neural Surrogates for HPC Applications:** Integrating machine learning with program analysis and compiler optimizations to accelerate performance-critical kernels in large-scale HPC codes, supported by the U.S. Department of Energy. (*To be submitted to IPDPS'27*)
Tech: C, C++, Fortran, Python, OpenMP, CUDA, ONNX, LLVM/Clang, LFortran, PyTorch, Numpy, TensorRT

- **Scaling Floating-Point Error Tracking with Whole-Program Shadow Execution:** Whole-program floating-point error tracking for optimized HPC applications using LLVM instrumentation and higher-precision shadow execution, with support for multi-module and MPI workloads. (*To be submitted to CGO'27*)
Tech: C, C++, Python, LLVM
- **Verifying Exact Conditions for Implementations of Density Functional Approximations:** An automated formal verification pipeline that proves scientific simulation code satisfies its required physical constraints, or produces concrete counterexamples where it does not, supported by the U.S. Department of Energy and NSF. (*To be submitted to CAV'27*)
Tech: C, C++, Python, dReal, SMT-LIB2
- **Dataflow Analysis of Event-Driven Programs:** Developed a static-analysis technique for event-driven programs (e.g., Android apps) by modeling them as multi-threaded programs, constructing Control-Vector Flow Graphs (CVFGs), and applying Sync-CFG-based points-to and null-dereference analysis, implemented in the tool **StAnDroid** ([\[Paper\]](#)).
Tech: Java, Soot, Android Studio
- **Building a Per-Process System-Call Sandbox:** Implemented a Linux kernel security module and static-analysis pipeline to extract and optimize system-call graphs as NFAs, integrate function-level graphs into a unified control automaton, and enforce fine-grained per-process system-call policies for enhanced runtime security.
Tech: C, Python, Angr, NetworkX
- **Face Recognition from Limited Data:** Implemented a manifold-matching-based face recognition approach for learning from limited samples, inspired by prior research.
Tech: Python, NumPy, OpenCV
- **Exploring Overlapping Communities in DBLP Citation Network:** Analyzed the DBLP citation network to detect overlapping and hierarchical research communities using the BIGCLAM model, ranked influential papers within communities, and evaluated scalability on a large real-world dataset.
Tech: Python, NetworkX, NumPy

HONORS AND LEADERSHIP

- Delivered 4 technical seminar presentations on research papers in neuro-symbolic learning and program analysis.
- Mentored two undergraduate (B.S.) students in research and career development: *Keena Vaslioff* (now a graduate student at UC Davis) and *Aditya Seth* (currently a Software Engineer at Chevron).
- Awarded a highly competitive national merit-based graduate scholarship by the Government of India (2021–2023).
- Awarded the **University Gold Medal** for earning the **top academic rank** in B.E. Information Technology (2021).